Electric Current

Electric current is a measure of the rate of flow of charge (electrons)

The charge flows when the circuit is complete.
Charge is measured in coulombs (C)

The rate of flow is the number of coulombs flowing per unit of time.

When 1 coulomb of charge flows in 1 second it is called an ampere (A)

Current is measured using an ammeter. Ammeters are connected in series with the source of electricity

An ammeter is connected from its positive terminal to the positive terminal of the source.

The usual current flowing in our homes is about 8 A

Voltage

Voltage is a measure of the electrical pressure which causes the electrons to flow in the circuit.

It is measured in volts (V)

A volt meter is used to measure the voltage.
A voltmeter is connected in parallel with the appliance to be measured.

A volt meter is connected from its positive terminal to the positive terminal of the source.

Some typical voltages are :

dry cell	**1.5 V**
house voltage	**240 V**
car battery	**12 V**
smoke alarm	**9 V**

Electricity is transmitted from the power stations at very high voltages ie. 66 000 V